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FinTech and financial inclusion in emerging markets: a bibliometric analysis and future research agenda

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Abstract

Purpose – The purpose of this article is to explore how FinTech is transforming financial inclusion in emerging markets through a detailed bibliometric analysis. The study identifies key research trends, themes and gaps, providing both theoretical insights and practical recommendations for policymakers and financial institutions. By integrating FinTech into established models of financial inclusion, the article highlights its potential to drive inclusive economic growth. Additionally, it proposes a future research agenda to address emerging challenges and opportunities, ensuring that the impact of FinTech on financial inclusion in developing regions is fully realized.

Design/methodology/approach – The study employs a bibliometric analysis to examine the existing literature on FinTech and financial inclusion in emerging markets. Using data from the Scopus database, the analysis focuses on identifying key trends, research themes and gaps within the field. The methodology includes performance analysis to determine the most prolific authors, institutions and countries, as well as science mapping to visualize the intellectual structure and thematic evolution. The combination of quantitative bibliometric techniques and qualitative content analysis provides a comprehensive overview of the research landscape and informs the proposed future research agenda.

Findings – The findings reveal that FinTech is significantly advancing financial inclusion in emerging markets, with research increasingly focusing on mobile banking, peer-to-peer lending and blockchain technologies. The analysis identifies China, the USA and the UK as leading contributors to this research. Key themes include the role of FinTech in reducing financial barriers and promoting economic development. However, gaps remain in understanding the long-term impacts of FinTech on financial stability and the specific needs of marginalized populations. The study highlights the need for more targeted research to fully leverage FinTech's potential in driving inclusive growth in developing regions.

Originality/value – This study provides a unique contribution by offering a comprehensive bibliometric analysis of FinTech's role in financial inclusion, specifically within the context of emerging markets. Unlike previous studies that focus on individual technologies or regions, this research systematically maps the global research landscape, identifying key trends, gaps and future research opportunities. The involvement of Dr Petterson Ozili, an expert from the Central Bank of Nigeria, adds significant value, ensuring the analysis is both relevant and informed by practical expertise. This paper serves as a valuable resource for researchers, policymakers and practitioners aiming to leverage FinTech for inclusive economic growth.

Keywords Emerging markets, Innovation, Bibliometric analysis, Financial inclusion, Banks/other depository institutions, Fintech

Paper type Literature review

1. Introduction

The advent of Financial Technology (FinTech) has fundamentally transformed the delivery of financial services, offering new opportunities to bridge gaps in access, particularly in emerging markets (Banna *et al.*, 2022; Khan and Khan, 2023). FinTech encompasses a wide range of innovations, including mobile banking, peer-to-peer lending, blockchain technology, and

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digital payment systems, all of which have the potential to overcome traditional financial barriers and reach underserved populations (Chang *et al*., 2004). Its significance lies in the ability to provide affordable, accessible, and user-friendly financial solutions to marginalized communities. This transformation is especially critical in regions where conventional banking infrastructure is underdeveloped, leaving a substantial portion of the population unbanked or underbanked (Jenik *et al*., 2020).

Several theoretical frameworks explain the relationship between FinTech development and expanded financial access. The diffusion of innovations theory (Rogers, 2003) highlights how financial institutions adopt and disseminate new technologies to enhance service accessibility across different social groups. The financial intermediation theory illustrates how FinTech disrupts traditional banking by introducing innovative and cost-effective mechanisms such as peer-to-peer lending and crowdfunded investments, thereby increasing the availability of financial services (Jarvis and Han, 2021; Gomber *et al*., 2018). Institutional theory underscores the influence of formal institutions (laws and regulations) and informal factors (cultural norms, traditions, and religious beliefs) on individuals' willingness to adopt technology for financial transactions (Holmes *et al*., 2013). Meanwhile, the resource-based view (RBV) explains how FinTech firms and financial institutions utilize unique organizational resources—such as technological capabilities and human capital—to create competitive advantages in advancing financial accessibility (Barney, 1991; Wang *et al*., 2021). The technology acceptance model (TAM) (Davis, 1989) provides further insight, asserting that perceived usefulness and ease of use significantly influence individuals' adoption of new financial technologies, particularly in low-income and digitally excluded regions. Additionally, the bottom of the pyramid (BoP) theory (Prahalad, 2010) suggests that providing affordable and accessible financial products to the world's poorest segments can be both profitable and socially beneficial.

FinTech companies play a pivotal role in targeting BoP markets with tailored financial products that address the specific challenges faced by underserved populations. Solutions such as microloans, mobile savings platforms, and pay-as-you-go insurance schemes provide crucial financial access to those previously excluded from formal financial systems (Carè *et al*., 2023). However, many FinTech firms operate on a dual approach, serving not only the BoP but also small businesses, middle-income consumers, and even high-net-worth individuals. For example, while mobile money platforms like M-Pesa were initially designed for the unbanked, they have since expanded their services to include business payments and credit facilities, demonstrating the scalability of FinTech models. This broader service strategy enhances financial sustainability, allowing firms to remain viable while continuing to support financially excluded communities (Arner *et al*., 2020). Achieving this balance between inclusivity and commercial viability ensures that digital financial solutions cater to a diverse range of users (Liang *et al*., 2024).

Despite the growing body of research on FinTech's role in expanding financial access, the existing literature lacks a comprehensive analysis of dominant themes, methodological approaches, and key findings within this domain. A deeper examination is necessary to understand the trajectory of research, identify unexplored areas, and highlight emerging topics that warrant further investigation. This study aims to fill this gap by conducting a systematic review of the literature, offering valuable insights for researchers, policymakers, and practitioners. Our objective is to analyze the state of research on FinTech in the fields of business, finance, and management, identifying current areas of focus while proposing directions for future studies. To achieve this, we employ bibliometric citation analysis, data visualization techniques, and content analysis (Bretas and Alon, 2021).

Examining FinTech's impact is essential given its potential to address global financial disparities. Expanding access to financial services is widely recognized as a catalyst for economic development and poverty reduction, enabling individuals to save, invest, and manage financial risks more effectively (Greenwood and Scharfstein, 2013). In emerging economies, where socio-economic inequalities are more pronounced, leveraging FinTech-

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driven solutions can drive significant improvements in economic stability, entrepreneurial activity, and overall household welfare.

Understanding the mechanisms through which FinTech expands financial access is equally crucial. This knowledge enables policymakers to design regulatory environments that foster sustainable innovation while mitigating risks. It also guides future research, ensuring that scholars address the most impactful and underexplored questions in the field.

To advance this understanding, the present study employs bibliometric analysis to offer both quantitative and qualitative insights into the evolution of research on FinTech-driven financial accessibility. This approach addresses the following research questions: (1) How does FinTech contribute to expanding financial access in business, management, and finance literature? (2) What are the key research streams in this domain? (3) What opportunities exist for future research?

By identifying areas that have garnered significant scholarly attention and highlighting existing gaps, our study ensures a balanced advancement of both theoretical and practical knowledge. The findings underscore that the impact of FinTech is particularly pronounced in emerging markets, where innovative financial solutions are most needed.

This study contributes to the literature by addressing several gaps. While prior research primarily focuses on developed and developing economies (Demir *et al.*, 2022; Emara, 2022), limited attention has been given to emerging markets (Junarsin *et al.*, 2023; Lyons *et al.*, 2022). Moreover, few studies provide a comprehensive review of the current state of research on FinTech and its role in expanding financial access. Our study fills this gap by systematically mapping the literature, identifying critical research streams, and proposing a detailed agenda for future inquiry.

The novelty of our contribution lies in its integrative perspective, which extends beyond previous studies by linking technological advancements with policy and regulatory considerations (Ozili and Syed, 2024). Unlike prior work that often examines FinTech applications in isolation, our study provides a holistic analysis, highlighting how FinTech innovations interact with institutional frameworks and socio-economic conditions. Additionally, this research contributes to the broader discourse on the role of technology in creating inclusive financial ecosystems that provide equitable access to essential financial services. By focusing on the unique challenges and opportunities in emerging economies, we extend existing frameworks and offer insights directly applicable to addressing financial exclusion in these contexts.

The remaining sections of the study are organized as follows. In section 2 we explain the proposed methodology and data used for the analysis. In section 3 we show the findings of bibliometric analysis and in section 4 we propose an agenda for future research obtained thanks to the content analysis. Section 5 reports conclusions of our study.

2. Materials and methods

This paper employs the bibliometric analysis method introduced by Pritchard (1969), which has become prominent in both quantitative and qualitative research analyses. Bibliometric analysis is a quantitative linguistic technique used to examine bibliometric data, revealing scientific development and changes through patterns and relationships within a specific scientific field (Wilczewski and Alon, 2023). This method, although not new, complements and extends traditional qualitative literature reviews (Ramos-Rodríguez and Ruíz-Navarro, 2004), providing a concise overview accompanied by a list of references (Block and Fisch, 2020). Bibliometric analysis has been widely applied in review studies (Kumar *et al.*, 2024). The evolution of citation databases, along with advanced statistical techniques and software for scientific mapping and visualization, has led to the increased acceptance of bibliometric analysis as a robust methodology for evaluating specific disciplines and themes (Liu *et al.*, 2023). Our study employs both thematic and bibliometric analysis to investigate the research developments and thematic evolution in public financial inclusion and fintech literature.

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The data for this study is sourced from the Scopus database, which is extensively used and known for its broad coverage, making it ideal for mapping research areas that are insufficiently covered by other databases (Rodrìguez-Lòpez *et al*., 2020). Scopus is particularly suited for profiling niche research fields that may otherwise be inadequately represented (Kivimaa and Kern, 2016).

Accordingly, we developed a research query that allowed us to include all relevant papers published in the field. We systematically search articles on the intersection of FinTech and financial inclusion using the following keywords: ("fintech" OR "financial technology") AND ("financial inclusion" OR "financial access" OR "inclusive finance"). We focus on articles and review papers written in English and published in business, management, accounting, economics, econometrics, and finance. The search used the Scopus database to cover the 2015 to April 2024 period, and it produced 737 articles. Figure 1 reports the steps which guided us to the selection of the final sample of 313 papers.

We build a dataset of articles and use two bibliometric methods to analyze it. First, we conduct a performance analysis to identify the most prolific countries, institutions, journals, and authors. We perform a science mapping using VOSviewer. This technique creates a network of articles based on shared references. It helps us uncover the underlying structure of major clusters or themes, forming the intellectual foundation of the field, as described by Mukherjee *et al*. (2022).

3. Findings

3.1 Trends

The literature on financial inclusion and FinTech has expanded since 2016, with several increasing articles published yearly (Figure 2). Since 2019, the growth has become exponential, with over 20 articles yearly. The most productive year is 2023, with 83 publications. The first four months of 2024 witnessed 81.

3.2 Most prolific contributors

Table 1 shows a hierarchy of scholarly output. China has the highest percentage of publications (22%), followed by the USA (16%) and the UK (15%). With 7% of the publications, the Renmin University of China tops the list of institutional production. Other Asian institutions, like Universiti Malaya, are highly prolific. The top journals in rankings are Journal of Risk and Financial Management and Resources Policy.

Table 2 reports the most prolific contributors, with Ozili from Central Bank of Nigeria ranking the highest.

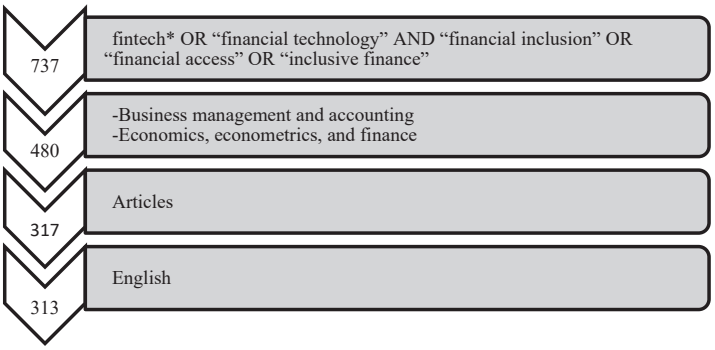

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Figure 1. A snapshot of articles search

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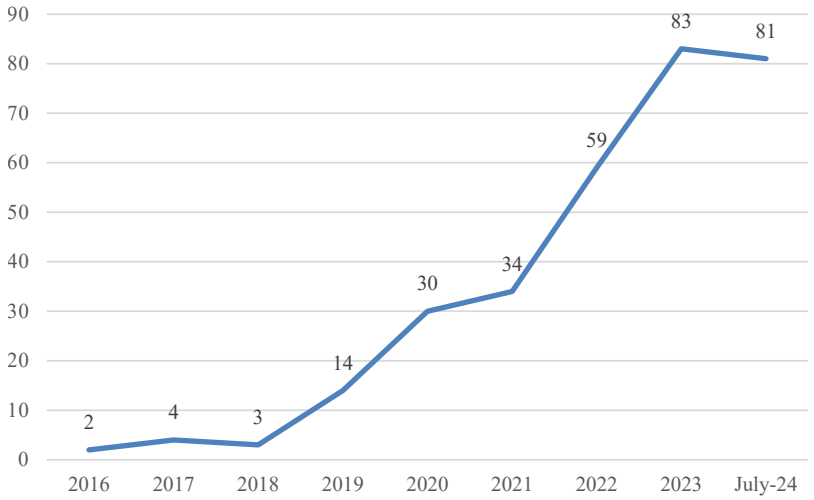

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Figure 2. Number of relevant publications per year

3.3 Most cited articles

Table 3 presents the most cited sources in FinTech and financial inclusion research, underscoring the most prominent themes. Junarsin *et al.* (2023) and Ozili (2021) examine FinTech's impact on financial stability and inclusion. Chen and Bellavitis (2020) discuss the disruption caused by blockchain technology and its decentralized finance models, like the comparative advantage of advertising expenditure research. Leong *et al.* (2017) explore business models within FinTech ecosystems, while Arner *et al.* (2020) study the role of FinTech in promoting sustainability, relating firm value and value creation to advertising literature. Demir *et al.* (2022) investigate the relationship between FinTech and income inequality, like research on mutual fund performance or systematic risk in stock performance.

3.4 Thematic map

The most significant topics related to fintech and financial inclusion are displayed on a two-dimensional strategic thematic map. This map illustrates the strength of internal links within a cluster (density, measured on the vertical axis) and the strength of external associations between different themes (centrality, measured on the horizontal axis). The positioning of themes within the four quadrants allows for a systematic analysis of their characteristics. These quadrants categorize themes into mainstream topics, basic and transversal topics, specialized topics, and emerging or underdeveloped topics, providing a comprehensive understanding of the thematic structure of the field (Bretas *et al.*, 2022).

Figure 3 shows a thematic map—an overview of the literary landscape. It unveils the clustering of current research and uncovers avenues for potential future exploration.

In the motor themes quadrant, central topics such as "financial markets," "sustainability," "economic growth," and "innovation" highlight key differences between developed and emerging markets. In developed countries, these themes focus on integrating advanced technologies like blockchain and green finance into mature financial systems, enhancing efficiency and promoting sustainable investments. In emerging markets, these themes play a transformative role, driving economic development and lifting populations out of poverty through innovations like mobile banking and digital payments. Sustainability efforts in these regions often align with broader development goals, such as improving energy access and climate resilience.

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Table 1. Most prolific countries, institutions, and journals

Most prolific countries				Most prolific institutions				Most prolific journals			
Rank	Country	Articles		Rank	Institutions	Articles		Rank	Journal	Articles	
		N	%			*N*	%			*N*	%
1	China	64	22	1	Renmin University of China	7	17	1	*Journal of Risk and Financial Management*	12	17
2	USA	45	16	2	Hungarian University of Agriculture and Life Sciences	5	12	2	*Resources Policy*	6	12
3	UK	43	15	= 2	Universiti Malaya	5	12	= 2	*Digital Policy, Regulation and Governance*	5	12
4	Indonesia	35	12	4	Multimedia University	4	10	4	*Journal of Open Innovation: Technology, Market, and Complexity*	5	10
5	India	33	11	= 4	Peking University	4	10	= 4	*Review of International Political Economy*	5	10
6	Malaysia	19	7	= 4	Universitas Indo Global MandirI	4	10	= 4	*Finance Research Letters*	4	10
7	Australia	18	6	7	Amity University	3	7	7	*Financial Innovation*	4	7
8	Hungary	11	4	= 7	Coventry University	3	7	= 7	*International Journal of Bank Marketing*	4	7
= 8	nigeria	11	4	= 7	Monash University	3	7	= 7	*International Journal of Social Economics*	4	7
= 8	United Arab Emirates	11	4	= 7	Nanyang Technological University	3	7	= 7	*Technological Forecasting and Social Change*	4	7

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Table 2. Most prolific contributors

Rank	Author	Country	Institution	Articles
1	Ozili	Nigeria	Central Bank of Nigeria	6
2	Banna	UK	Manchester Metropolitan University	5
3	Setiawan	Indonesia	Indo Global Mandiri University	4
4	Chen	China	Chinese Labor Research Institute	3
= 4	Fekete-Farkas	Hungary	Hungarian University of Agriculture and Life Sciences	3
= 4	Nathan	Malaysia	Multimedia University	3
7	Aduba	Japan	Ritsumeikan Graduate School of Economics	2
= 7	Al-okaily	Giordan	Associate Professor of Accounting Information Systems	2
= 7	Alam	Australia	The University of Queensland	2

Source(s): Authors' own work

Table 3. Most cited articles

Rank	Article	Authors	Journal	TC	C/Y
1	Impact of digital finance on financial inclusion and stability	Ozili (2018)	*Borsa Istanbul Review*	641	91.57
2	Blockchain disruption and decentralized finance: The rise of decentralized business models	Chen and Bellavitis (2020)	*Journal of Business Venturing Insights*	298	59.60
3	Nurturing a FinTech ecosystem: The case of a youth microloan startup in China	Leong *et al.* (2017)	*International Journal of Information Management*	208	26.00
4	Sustainability, FinTech and financial inclusion	Zetzsche *et al.* (2017)	*European Business Organization Law Review*	199	39.80
5	Fintech, financial inclusion and income inequality: a quantile regression approach	Demir *et al.* (2022)	*The European Journal of Finance*	191	63.67
6	Industry 4.0 in finance: the impact of artificial intelligence (ai) on digital financial inclusion	Mhlanga (2020)	*International Journal of Financial Studies*	174	34.80
7	Unearthing antecedents to financial inclusion through FinTech innovations	Senyo and Osabutey (2020)	*Technovation*	159	31.80
8	Blockchain entrepreneurship opportunity in the practices of the unbanked	Larios-Hernández (2017)	*Business Horizons*	135	16.88
9	Financial inclusion research around the world: A review	Ozili (2021)	*Forum for Social Economics, Routledge*	126	31.50
10	From fintech to finlife: The case of fintech development in China	Chen *et al.* (2023)	*China Economic Journal*	115	12.78

Source(s): Authors' own work

Niche themes like "green finance" and "finance system" are well-developed but less central. In developed markets, green finance supports the transition to low-carbon economies through advanced financial instruments and regulatory frameworks. In emerging markets, it focuses on attracting sustainable investments and supporting renewable energy and SMEs, while simultaneously building basic financial infrastructure.

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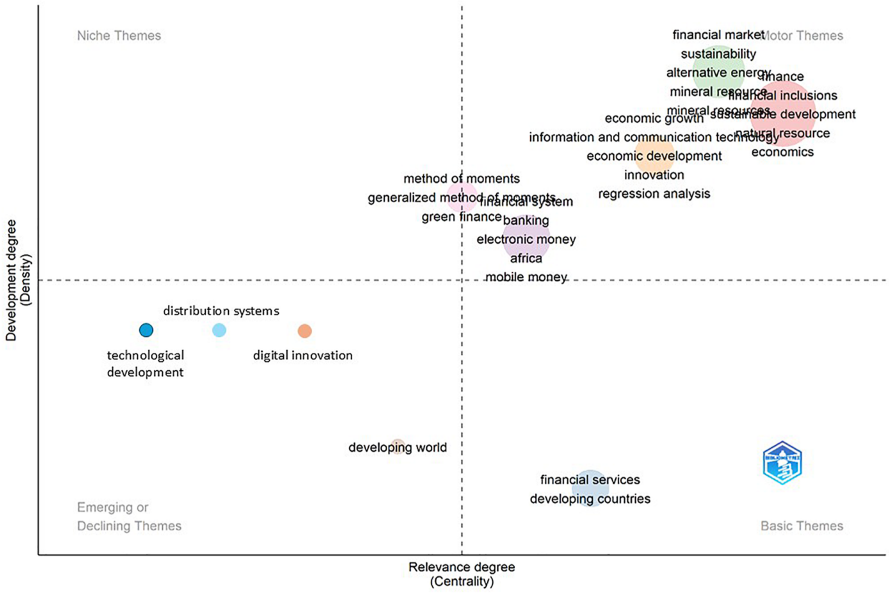

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Figure 3. Thematic map

Emerging or declining themes such as "technological development" and "digital innovation" hold significant potential in emerging markets, where technology bridges infrastructure gaps and provides financial access to underserved populations, exemplified by platforms like M-Pesa. In developed markets, these themes refine existing technologies, improving efficiency and customer experiences with innovations like blockchain and AI.

"Financial services" and "developing countries" are basic but underdeveloped themes. In emerging markets, the focus is on expanding access to basic banking and credit services, addressing barriers like high costs and regulatory challenges. In developed countries, advanced technologies drive diversification, offering solutions like robo-advisors and personalized banking services to meet sophisticated consumer needs.

The thematic map illustrates distinct contrasts between emerging and developed markets. Emerging markets prioritize building inclusive financial systems through scalable, affordable solutions that address unbanked populations. Developed markets, in contrast, optimize advanced systems, focusing on sustainability, innovation, and maintaining global competitiveness.

While both contexts aim to advance financial inclusion and economic development, their approaches differ significantly. Emerging markets focus on foundational growth through innovation, whereas developed countries enhance existing systems for efficiency and sophistication. This analysis underscores the need for context-specific strategies to foster inclusive and resilient financial systems globally.

3.5 Bibliographic coupling

We conduct a bibliographic coupling analysis on highly cited articles in FinTech and financial inclusion to identify the main clusters or themes. Previous studies show that bibliographic coupling techniques can be used to identify "hot" research topics and emerging research trends (Jain *et al*., 2022). Therefore, the second step of the analysis was to perform Bibliographic

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Coupling Analysis (BcA). This approach allows us to detect clusters of research that are gaining momentum, highlighting the areas that are currently attracting significant scholarly attention (Kumar *et al*., 2024). By identifying these hot topics and emerging lines of research, we can guide future investigations towards the most promising and impactful areas, ensuring that subsequent studies are aligned with the latest developments and gaps in the field (García-Lillo *et al*., 2021).

Figure 4 presents the results of the bibliographic coupling analysis. Each node in the diagram represents an author or group of authors (if they have co-published), and the size of each node reflects the number of publications or influence of that author or group within the field. Larger nodes, such as the one for Ozili (2018), indicate more extensive publications or higher influence. The edges connecting the nodes represent the degree of bibliographic coupling, with thicker edges indicating stronger connections based on shared citations in their published works. For instance, the thick edge between Ozili (2018) and Emara (2022) suggests a high degree of shared references, indicating thematic alignment or influence between these authors.

This analysis grouped the articles into four major clusters, as depicted in Figure 4. Each cluster highlights a distinct thematic area gaining traction in the field of FinTech and financial inclusion. Before diving into the analysis of these clusters, we provide an overview of their characteristics in Figure 4, which serves as a foundation for understanding the relationships and trends among the authors and topics included in this study.

3.5.1 Cluster 1: yellow. 3.5.1.1 Bridging financial gaps for inclusive growth. Financial technology (FinTech) has emerged as a key catalyst for expanding access to financial services and promoting sustainable development, as evidenced by recent studies.

Demir *et al*. (2022), analyzing Global Findex data across 140 countries, reveal that FinTech influences income inequality both directly and indirectly by broadening financial access. Their findings indicate that greater financial accessibility significantly reduces inequality, particularly in higher-income nations, aligning with the Agenda for Sustainable Development and the G20 High-Level Principles for Digital Financial Inclusion. Similarly, Arner *et al*. (2020) emphasize FinTech's role in advancing the Sustainable Development Goals

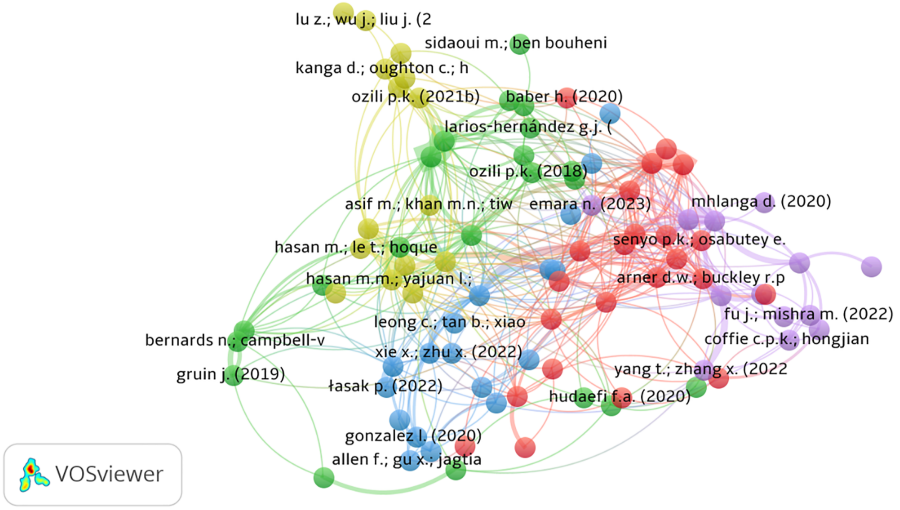

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Figure 4. Bibliographic coupling

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(SDGs) through a four-pillar strategy: digital identities, interoperable payment systems, electronic government services, and digital financial markets. This framework underscores FinTech's potential to reshape economies by fostering more inclusive and resilient financial ecosystems.

In developing countries, Al-Okaily (2023) identifies key determinants of e-loyalty towards digital wallets, including perceived trust, financial cost, and ease of use, based on a study of Jordanian users. These findings provide policymakers with valuable insights into encouraging widespread adoption of digital payment solutions. Complementing this, Senyo and Osabutey (2020) investigate behavioral factors influencing FinTech usage, particularly mobile money. Utilizing UTAUT2 and Prospect Theory, they find that performance and effort expectancy significantly drive adoption, while price value and perceived risk have a lesser impact. Their research sheds light on consumer attitudes toward digital financial tools.

In the UAE, Aloulou *et al.* (2024) examines FinTech's impact on the banking sector during COVID-19, revealing that technological advancements have strengthened competitiveness and overall performance. Effective technology management emerges as a crucial factor in maintaining an edge within this dynamic financial landscape.

Together, these studies illustrate the profound impact of FinTech in expanding economic participation, reducing inequality, and fostering sustainable growth, reinforcing its critical role in achieving global development objectives.

3.5.2 Cluster 2: green. 3.5.2.1 Technological advancements and financial stability. This cluster explores the impact of FinTech innovations on expanding access to financial services and ensuring systemic stability, particularly in developing and emerging economies. Banna *et al.* (2021) demonstrate that greater adoption of FinTech-driven financial solutions can moderate banks' risk-taking behavior, though the competitive pressures introduced by digital financial services also contribute to increased uncertainty. This dual effect highlights the need for regulatory frameworks that balance technological advancements with financial system resilience.

Bernards and Campbell-Verduyn (2019) emphasize the disruptive nature of FinTech, advocating for a nuanced understanding of how technological changes interact with existing financial structures. Their heuristic approach integrates perspectives from International Political Economy, Science and Technology Studies, and Social Studies of Finance, offering a comprehensive framework for assessing the broader implications of digital finance.

Banna *et al.* (2022) extend this analysis to microfinance institutions (MFIs) in Sub-Saharan Africa, developing an index to measure the impact of FinTech on financial accessibility. Their findings suggest that digital financial tools enhance MFI stability by mitigating risk-taking behaviors, particularly in regions with underdeveloped banking infrastructures. This research underscores how FinTech can contribute to financial system resilience by supporting risk management within alternative lending institutions.

Brandl and Dieterich (2023) examine the digital transformation of global payment systems, revealing that while technology-driven firms are reshaping the sector, reliance on centralized trust mechanisms persists and often increasing operational complexity. Their study underscores the necessity of regulatory oversight to ensure stability within evolving financial networks.

Ozili (2018) explores the benefits and risks of digital finance in developing markets, noting its potential to expand financial access while also identifying challenges such as inadequate consumer protection and technological vulnerabilities. His research highlights the importance of considering socio-economic contexts and implementing holistic regulatory approaches to maximize the benefits of digital financial solutions.

In summary, this cluster underscores the intricate relationship between FinTech-driven financial access and systemic stability. The studies emphasize the need for a deeper understanding of digital financial infrastructures, robust regulatory oversight, and proactive strategies to mitigate emerging risks. While FinTech presents significant opportunities for

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fostering inclusive and sustainable financial development, its transformative potential must be carefully managed to ensure long-term stability.

3.5.3 Cluster 3: blue. 3.5.3.1 The drivers of financial inclusion and sustainability. This cluster highlights the transformative role of financial technology (FinTech), decentralized finance, and information and communication technologies (ICTs) in advancing financial inclusion, sustainability, and poverty alleviation globally.

Allen *et al*. (2022) emphasize FinTech's penetration into all areas of the financial system, particularly through cryptocurrencies, stablecoins, and central bank digital currencies (CBDCs). The study focuses on China's progress, highlighting how effective regulation fosters innovation and builds trust, with the potential for the e-CNY to become a mainstream digital currency. Leong *et al*. (2017) examine FinTech's impact on traditional financial sectors, showcasing how digital tools enable niche markets, alternative credit scoring, and financial inclusion, using the example of microloans for college students in China.

Liu *et al*. (2023) explore ICTs' role in enhancing microfinance institutions' efficiency and sustainability through innovations like crowdfunding, P2P lending, and mobile banking. The study predicts that technologies such as blockchain will continue to drive financial inclusion and entrepreneurship. Ofosu-Mensah Ababio *et al*. (2023) link financial inclusion to renewable energy adoption in developing countries, showing that improved access to finance supports green technology and sustainability initiatives. Ali and Marisetty (2023) analyze global poverty alleviation through FinTech and microfinance, identifying key trends in crowdfunding, financial literacy, and their intersection with natural resource management and economic growth.

In summary, this cluster underscores how FinTech and ICTs revolutionize financial services, promote inclusion, and address global challenges like poverty and environmental sustainability. The studies highlight the critical role of regulation, digital innovation, and targeted financial strategies in leveraging technology for sustainable socio-economic development.

3.5.4 Cluster 4: red. 3.5.4.1 Financial inclusion and economic growth: contrasts between emerging and developed markets. This cluster of articles focuses on the pivotal role of financial inclusion in fostering economic stability and growth, particularly in emerging markets, while highlighting the contrasts with developed markets.

Higher financial inclusion enhances bank stability, especially in banks with higher customer deposit funding and lower marginal costs and is more pronounced in countries with strong institutional quality (Ahamed and Mallick, 2019). This finding suggests that emerging markets, often characterized by weaker institutional frameworks, could see significant gains in financial stability through targeted inclusion policies (Pahlevan Sharif *et al*., 2023).

Financial inclusion brings substantial benefits to typically excluded groups such as the poor, rural, female, or young, with effective policies including reducing account costs, increasing proximity to services, ensuring stronger legal rights, and maintaining political stability (Allen *et al*., 2016). These policies are particularly crucial in emerging markets where financial access barriers are more pronounced, contrasting with developed markets where financial infrastructure is more established.

Mobile phone diffusion significantly contributes to economic growth in African countries by enhancing financial inclusion, underscoring the importance of technology in bridging financial infrastructure gaps (Andrianaivo and Kpodar, 2012). This highlights a key divergence between emerging and developed markets, where the former can leapfrog traditional financial systems through mobile technology. Economic development is modeled through institutional transformation influenced by wealth distribution and capital market imperfections, demonstrating long-term effects based on initial wealth distributions (Banerjee and Newman, 1994). This underscores the need for policies that address initial inequalities to ensure sustainable development, a challenge more acute in emerging markets.

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Financial development significantly benefits the poorest quintile, reducing income inequality and boosting incomes, emphasizing the crucial role of the financial system in aiding the poor and fostering inclusive economic progress (Beck *et al.*, 2007). This is particularly relevant for emerging markets, where income inequality is often higher and financial systems less developed compared to developed markets. Collectively, the cluster underscores that financial inclusion and development are instrumental in driving economic stability and growth, with tailored policies needed to address specific demographic needs in different markets. Emerging markets, with their distinct financial infrastructure challenges, particularly benefit from innovations like mobile banking to bridge gaps and support inclusive economic development. Conversely, developed markets may focus on refining institutional quality and reducing service costs to further enhance financial inclusion. Thus, the cluster highlights the multifaceted benefits of financial inclusion across different economic contexts and the importance of context-specific policy interventions.

3.5.5 Cluster 5: violet. 3.5.5.1 Transformative fintech: enhancing financial inclusion in emerging markets. The cluster of articles revolves around the transformative role of fintech in enhancing financial inclusion, particularly in emerging markets, by addressing barriers such as geographical limitations, high costs, inappropriate banking products, and financial illiteracy. Gabor and Brooks (2017) highlight how digital ecosystems, driven by fintech, philanthropic investments, and state institutions, expand financial access through behavioral economics and predictive algorithms, thus enabling states and global finance to integrate excluded households into formal financial systems. Schuetz and Venkatesh (2020) propose blockchain technology as a solution to financial exclusion in rural India, emphasizing that overcoming challenges related to geographical access, cost, product suitability, and financial literacy can connect rural populations to global supply chains, thereby driving economic development. Bernards and Campbell-Verduyn (2019) critically examines the uneven and limited rollout of fintech, arguing that the reductive nature of credit scoring systems and the precarious livelihoods of users often constrain fintech's reach, prompting a need for realistic assessments of its capabilities. Gomber *et al*. (2018) discuss the disruptive innovations brought by fintech, mapping technological advancements in payments, cryptocurrencies, blockchain, and lending, and highlighting the challenges faced by traditional financial institutions in adapting to these changes, which are essential for improving efficiency, customer centricity, and service transformation. Lee *et al.* (2022) provide a historical overview of fintech, exploring different business models and the technical and managerial challenges faced by both startups and traditional institutions, emphasizing the strategic importance of understanding fintech investments. Collectively, these articles underscore fintech's potential to revolutionize financial inclusion in emerging markets by offering innovative solutions that bridge existing financial infrastructure gaps, while also acknowledging the need for tailored, context-specific approaches to address practical challenges and limitations. The studies emphasize that for fintech to realize its full potential, there must be concerted efforts to adapt technologies to local contexts, ensure regulatory support, and address socio-economic disparities, which are more pronounced in emerging markets compared to developed ones.

4. An agenda for future research

The literature on the impact of fintech on financial inclusion is rapidly expanding. Nevertheless, additional studies are necessary to address fundamental questions and explore related concepts. This study seeks to propose a future research agenda based on trends identified through a content analysis of 313 papers. Table 4 summarizes our recommended future research questions.

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Table 4. Future research questions

Research stream	Focus	Future research questions
Fintech for financial inclusion in emerging markets	Exploring how FinTech solutions contribute to financial inclusion in emerging markets, where access to traditional financial services is often limited	How do FinTech innovations improve access to financial services for unbanked and underbanked populations in emerging markets? What are the most successful FinTech models and technologies being implemented in emerging markets, and what factors contribute to their success? What are the roles of mobile money, digital wallets, and peer-to-peer lending platforms in enhancing financial inclusion in these regions? How can FinTech companies collaborate with governments, NGOs, and local communities to ensure that their solutions are sustainable and impactful?
Gender and financial inclusion via FinTech	Understanding the gender-specific impacts of FinTech on financial inclusion, particularly how FinTech services are addressing the financial needs of women	How does FinTech address the unique financial challenges faced by women in emerging markets? What are the successful case studies of FinTech improving women's access to financial services and economic opportunities? How do cultural and social factors influence the adoption of FinTech solutions among women?
FinTech for rural development	Assessing the impact of FinTech solutions on financial inclusion in rural areas, where traditional banking services are often limited	How do FinTech innovations bridge the gap between urban and rural financial services, and what models have proven most effective? What are the successful strategies for implementing FinTech in rural communities, considering factors such as digital literacy, infrastructure, and cultural acceptance? What partnerships between FinTech companies, governments, and local organizations are necessary to enhance the reach and impact of FinTech in rural areas? How can FinTech solutions be designed to meet the specific needs of rural populations, including smallholder farmers, rural entrepreneurs, and low-income households?
Fintech for sustainable development	Exploring how fintech can drive sustainable development by promoting financial inclusion, supporting green financing, and facilitating capital access for responsible projects	How can Fintech innovations contribute to the achievement of the United Nations Sustainable Development Goals (SDGs) by enhancing financial inclusion in developing countries? What role does blockchain technology play in promoting transparency and accountability in sustainable finance initiatives? How can Fintech be utilized to create scalable models for sustainable investment in renewable energy projects in emerging markets?

(*continued*)

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Table 4. Continued

Research stream	Focus	Future research questions
Synergy between Fintech and artificial intelligence	Exploring how AI integration enhances fintech by driving innovation in personalized services, risk assessment, and automated decision-making	How can the integration of Artificial Intelligence (AI) in Fintech platforms enhance predictive analytics for better risk management in financial services? What are the ethical implications of deploying AI-driven Fintech solutions in consumer financial services, and how can these be addressed to ensure fairness and transparency? How can AI and Fintech jointly create personalized financial products that cater to the unique needs of underbanked populations?
Fintech for climate change mitigation	Exploring how fintech solutions can support efforts to combat climate change by enabling green financing, promoting carbon offset initiatives, and facilitating investments in sustainable projects	How can Fintech innovations enable more effective climate risk assessments and adaptation strategies for businesses and governments? What is the impact of digital financial platforms on accelerating the adoption of sustainable practices among small and medium-sized enterprises (SMEs) in high-risk climate zones? How can peer-to-peer (P2P) lending platforms powered by Fintech contribute to funding grassroots climate change mitigation efforts?

Source(s): Authors' own work

4.1 Fintech for financial inclusion in emerging markets

The rapid advancement of financial technology (fintech) has revolutionized financial services, particularly in emerging markets where traditional banking infrastructure is often lacking. Fintech solutions, such as mobile banking, peer-to-peer lending, and blockchain-based transactions, have significantly expanded access to financial services for underserved populations. These innovations enable individuals and small businesses to participate in the formal economy by providing easier access to credit, secure methods of saving, and efficient payment systems. For instance, mobile money platforms like M-Pesa in Kenya have demonstrated substantial success in enhancing financial inclusion by allowing users to transfer money, pay bills, and save securely using their mobile phones. Moreover, fintech has the potential to reduce transaction costs, improve transparency, and increase the overall efficiency of financial systems, thus fostering economic growth and reducing poverty in these regions. As emerging markets continue to adopt and integrate fintech solutions, it is essential to address regulatory challenges and ensure that these technologies are accessible and beneficial to all segments of the population.

4.2 Gender and financial inclusion via FinTech

FinTech has become a transformative tool for advancing gender financial inclusion, particularly in regions where traditional banking has marginalized women. Mobile money platforms like M-Pesa in Kenya and bKash in Bangladesh enable women to save, transfer money, and access credit without needing a physical bank account, addressing logistical and societal barriers. Innovations like digital credit scoring, which use alternative data such as mobile phone usage and utility payments, allow women to obtain loans despite lacking formal

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credit histories. Platforms like Tala, operating in Kenya and the Philippines, exemplify how these advancements empower women economically, fostering entrepreneurship and financial independence.

These developments contribute to broader economic growth, with improved household welfare and increased investment in education and health. However, challenges remain, including digital literacy gaps and unequal access to mobile devices and the internet. Addressing these barriers through training programs and community initiatives is essential to maximize FinTech's impact. Tailored products, such as flexible savings accounts or microinsurance for women entrepreneurs, can further enhance inclusion.

Regional examples illustrate the diverse impact of FinTech. While mobile money leads in Sub-Saharan Africa, microfinance-linked digital solutions are prevalent in South Asia. Case studies, such as India's Jan Dhan-Aadhaar-Mobile initiative, highlight how combining technological innovation with supportive policies can significantly bridge gender gaps in financial systems.

4.3 FinTech for rural development

FinTech plays a pivotal role in fostering rural development by bridging the gap between rural populations and essential financial services. Traditional banking infrastructure often fails to reach remote and rural areas, leaving residents without access to savings accounts, credit facilities, and payment systems. FinTech innovations, such as mobile banking, microfinance platforms, and blockchain-based solutions, provide these underserved communities with critical financial tools. Mobile banking, for example, allows individuals to conduct transactions, save money, and access credit through their mobile phones, circumventing the need for physical bank branches. Microfinance platforms enable small farmers and entrepreneurs to secure loans with minimal requirements, fueling local economic activities and improving livelihoods. Blockchain technology can enhance transparency and reduce transaction costs in agricultural supply chains, ensuring fair prices for rural producers. These technologies not only facilitate financial inclusion but also promote economic resilience and growth in rural areas. However, to maximize the benefits of FinTech for rural development, it is crucial to invest in digital literacy programs and ensure affordable access to mobile devices and internet connectivity.

4.4 Fintech for sustainable development

The exploration of fintech's role in sustainable development offers a promising avenue to deepen our understanding of how financial innovation can contribute to global sustainability goals. Future research should focus on identifying the mechanisms through which fintech can enhance financial inclusion and support green financing initiatives, particularly in underserved regions. Additionally, investigating the regulatory frameworks required to balance the rapid growth of fintech with the protection of environmental and social governance (ESG) standards is crucial. Examining the long-term impacts of fintech on economic development and environmental sustainability is essential to assess both the opportunities and potential risks. Addressing these aspects will enhance our understanding of fintech's potential to drive sustainable development and create strategies that align technological advancement with the achievement of global sustainability objectives.

4.5 Synergy between fintech and artificial intelligence

The intersection of fintech and artificial intelligence (AI) represents a powerful synergy with the potential to revolutionize the financial industry. AI-driven technologies can enhance fintech solutions by enabling more accurate risk assessment, personalized financial services, and improved decision-making processes. By integrating AI into fintech platforms, companies can automate complex financial tasks, such as fraud detection and credit scoring, while also

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providing tailored financial advice to consumers based on real-time data analysis. Furthermore, the combination of fintech and AI can significantly increase operational efficiency and reduce costs for financial institutions. Exploring the synergy between these two fields not only highlights the transformative potential of their convergence but also underscores the need for robust ethical frameworks to ensure that advancements in AI-driven fintech contribute to fair and inclusive financial systems.

4.6 Fintech for climate change mitigation

Fintech has the potential to play a pivotal role in climate change mitigation by enabling innovative financial solutions that support environmentally sustainable practices. Through the use of digital platforms, fintech can facilitate the mobilization of capital towards green projects, such as renewable energy initiatives and carbon offset programs. Additionally, fintech can enhance transparency and accountability in carbon markets by utilizing blockchain technology to track emissions and ensure compliance with environmental standards. The development of green bonds and climate-focused investment funds further exemplifies how fintech can drive funding towards sustainable ventures. By leveraging these technologies, fintech can significantly contribute to global efforts in reducing carbon footprints and promoting a low-carbon economy, making it a key player in the fight against climate change.

5. Discussion and conclusion

This article presents a comprehensive analysis of the role of FinTech in promoting financial inclusion, with a particular focus on emerging markets, where traditional banking infrastructure remains weak and financial exclusion is most acute. Through a bibliometric study, we mapped the evolution of research in this field, identifying key trends, dominant themes, and gaps that require further exploration. By integrating empirical case studies from India, Kenya, and Brazil, we illustrate how FinTech solutions such as mobile banking, alternative credit scoring, and digital payment platforms have transformed access to financial services for underserved populations, reducing systemic barriers and fostering economic empowerment. Additionally, we critically examine the theoretical frameworks that explain the adoption and diffusion of these technologies, offering a research agenda that addresses pressing questions at the intersection of FinTech and financial inclusion.

This study makes significant contributions to the literature by bridging theoretical insights with empirical evidence and addressing critical gaps in existing research. Prior bibliometric studies, including those by Vasishta *et al*. (2024), Kishor *et al*. (2024), and Afjal (2023), have explored trends in FinTech and financial inclusion, yet they often provide broad overviews or focus on specific regions without fully analyzing the mechanisms through which FinTech mitigates systemic barriers. In contrast, our study delves into the socio-economic contexts where financial exclusion persists, highlighting how FinTech solutions not only disrupt traditional financial systems but also introduce targeted innovations that promote equity and sustainable development. For example, M-Pesa in Kenya has enabled millions of unbanked individuals to engage in financial transactions without reliance on conventional banking infrastructure, while AI-driven credit scoring models in Brazil have expanded access to capital for micro-entrepreneurs traditionally excluded from formal financial institutions. These cases underscore the transformative potential of FinTech in bridging financial gaps, particularly in environments characterized by regulatory and infrastructural constraints.

A key aspect of our contribution lies in the integration of established theoretical models with empirical insights, offering a more nuanced understanding of FinTech adoption and diffusion. While previous research has largely relied on the Diffusion of Innovations theory and the Technology Acceptance Model (TAM) (Chan *et al*., 2022; Singh *et al*., 2020), our study extends these frameworks by incorporating Institutional Theory and the Resource-Based

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View (RBV). Institutional Theory provides insights into how regulatory environments and cultural norms shape the pace and nature of FinTech adoption, as evidenced by the divergent trajectories of mobile money adoption in Kenya and Mexico, where differing regulatory landscapes have either facilitated or constrained innovation. The RBV framework further explains how FinTech firms leverage technological resources, data analytics, and proprietary algorithms to create sustainable competitive advantages in financial inclusion. The Aadhaar-based digital identity system in India exemplifies this approach, as financial institutions utilize biometric authentication to facilitate access to banking services for previously excluded populations (Ozili and Syed, 2024). By synthesizing these theoretical perspectives with real-world cases, our study presents a more holistic understanding of how FinTech operates at the intersection of innovation, regulation, and socio-economic development.

Beyond its theoretical contributions, this study provides actionable insights for policymakers, financial institutions, and technology developers, offering concrete strategies to enhance FinTech-driven financial inclusion. A key finding underscores the need for regulatory frameworks that balance innovation with consumer protection, particularly in emerging markets where financial oversight remains fragmented. Regulatory sandboxes, as successfully implemented in Singapore and the UAE, serve as effective models for enabling FinTech experimentation under controlled conditions while ensuring financial stability. The study also emphasizes the importance of tailoring FinTech solutions to marginalized groups, including women, rural populations, and informal sector workers. Targeted financial education programs, as demonstrated by Bangladesh's bKash and Indonesia's GoPay, play a crucial role in increasing user trust and engagement. Additionally, alternative credit scoring models leveraging AI and mobile data can expand access to capital for individuals lacking traditional credit histories. A further contribution is highlighting the role of public-private partnerships (PPPs) in scaling financial inclusion initiatives. Successful collaborations between governments, FinTech firms, and development organizations – such as in sub-Saharan Africa – have enhanced financial infrastructure and expanded access to digital financial services.

By bridging FinTech innovation with practical applications, this study provides evidence-based recommendations for regulatory policies, product design, and strategic investments, ensuring that digital financial solutions drive equitable and sustainable financial inclusion.

A unique aspect of this study is its ability to bridge bibliometric analysis with practical applications, ensuring that the insights derived from research trends translate into meaningful policy and business strategies. While previous bibliometric studies, such as those by Afjal (2023) and Chhikara (2022), have primarily documented trends in FinTech adoption, our research extends this approach by linking emerging research clusters with concrete recommendations for academia and industry. Key policy imperatives that emerge from our analysis include expanding digital infrastructure to enhance mobile banking penetration, encouraging inclusive regulatory frameworks that support financial innovation, developing consumer protection mechanisms to address data security risks, and investing in financial literacy programs to empower end users. These strategic interventions are critical in ensuring that FinTech solutions are not only widely accessible but also equitable and sustainable in the long run.

In conclusion, this study provides a comprehensive and integrative analysis of FinTech's role in financial inclusion, with a particular focus on emerging markets where financial exclusion remains a pressing issue. By advancing theoretical frameworks, incorporating empirical case studies, offering policy-relevant insights, and bridging bibliometric research with practical applications, our work makes a significant contribution to the academic discourse and provides a roadmap for future investigations. The findings highlight the importance of interdisciplinary approaches and call for a deeper exploration of the long-term impact of FinTech innovations on financial inclusion, particularly in underserved regions. As financial ecosystems continue to evolve, research that integrates empirical evidence with theoretical rigor will be essential for addressing the complex and dynamic challenges of financial exclusion in the digital age.

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